Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/23/2005
PI Name:	Lipa, John A. Ph.D.		
Project Title:	Universality Tests on DYNAMAX		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	04/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 04/30/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2873		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity.		

Task Description:	We had originally planned to make use of the DYNAMX apparatus to be flown in the LTMPF to measure the exponent characterizing the divergence of the superfluid density near the lambda point of helium. This quantity is of great theoretical interest for testing modern theories of phase transitions in condensed matter. Due to the cancellation of the FP discipline, work has been diverted into related ground measurements involving the specific heat and cnnvesion of earlier data to superfluid density with higher accuracy. The DYNAMX experiment provides access to the lambda transition of helium with the effects of gravity removed. This allows the testing of various aspects of phase transition theory with unprecedented accuracy. Our measurements will probe the behavior of the superfluid density and the heat capacity very near the transition to the level possible on the ground. The results of the measurements will be used to determine the behavior of the heat capacity and the superfluid density very near the transition. This will be a new test of modern theories of phase transitions in condensed matter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved understanding of the applicability of renormalization techniques could affect the scientific infrastructure underlying many areas of practical value, from the limits on miniaturization of computer chips to the development of new types of polymers.
Task Progress:	In the past year we have focused on collecting new heat capacity data along the lambda line of helium to serve as precursor measurements for the potential flight experiment using the DYNAMX instrument. Accurate measurements of the specific heat of helium near the lambda line as a function of pressure are important for testing the predictions of universality that are a central feature of the theory of second order phase transitions. Universality predicts that the exponents characterizing the divergence of the specific heat and the singular behavior of the superfluid density are independent of pressure, P, or 3He concentration along the respective lambda lines, and that some ratios of coefficients show similar behavior. These predictions have been tested with moderate precision in a number of experiments and some small discrepancies have been noted. Over the past few years we have built a new apparatus for measuring of the constant-volume specific heat, Cv, in which the fluid pressure can also be recorded as a function of temperature. This information enables us to compute the constant-pressure specific heat, Cp, to which universality should apply. It will also allow us to compute the superfluid density from earlier second sound velocity measurements more accurately than has been possible to date. The present results extend significantly closer to the transition than previous work, allowing a better determination of the asymptotic behavior. So far we have collected data and performed a preliminary analysis of the measurements using earlier pressure measurements obtained by G. Ahlers to convert from Cv to Cp. Our own pressure measurements have not yet been analyzed, but these are expected to result in significantly more accurate Cp data close to the lambda point.
Bibliography Type:	Description: (Last Updated: 11/02/2007)
Papers from Meeting Proceedings	Lipa JA, Nissen JA, Avaloff D, Wang S. "Specific heat of helium at constant volume along the lambda line." LT24 meeting, 24th International Conference on Low Temperature Physics, Orlando, FL, August 10-17, 2005. LT24 meeting, 24th International Conference on Low Temperature Physics, Proceedings in press, August 2005. , Aug-2005